

* Memory Management in Variable

RAM (Random Access Memory)

Stack Memory

(It is used to store variable names / pointers)

(It is stored in organised way orderly placed)

Mainframe

(It is used for execution purpose)

Heap Memory

(It is used to store objects / Value)

(It stored randomly unorganised way)

reference

ref count = 0

10

9

b

c

20

30

ref count = 0

garbage collection.

— The reference count of 10 is 0

— The reference count of 20 is 2.

— Reference count of 30 is 1.

Theory :-

When we want to execute my code in python RAM is required.

Variables starts pointing towards respective values and reference count of each value will be 1.

Since python supports dynamic type memory allocation & we can change the value for variables whenever we required.

we will deference to from variable 'a' & starts pointing towards variable 'b'. Now, reference count of 20 becomes 2 and 10 becomes 0, because no variables pointing towards 10.

When reference count becomes zero (0) the garbage collection comes into picture and remove value from memory and free up space and this is how memory management happens in variables.

* IDENTIFIERS

Python program
It is name given to variables, function, class, modules and other objects.

If you are assigning some name to programmable entity in python then it is nothing but technically called an identifiers.

Rules:-

1) In identifier part we can pass both alphabets (upper case / lower case / combination)

abc = 10

abc

o/p = 10

2) In identifier part we can use number but should not start with number. We can use number in between or last

eg: a34 = "hi"

3) Special symbols are not allowed except underscore (_)

eg: @ = 90

SyntaxError: invalid syntax

4) Python documents says that you can pass on identifier with unlimited length. (maximum 79 characters)

5) keywords can not be used in identifier part

eg: in = 89

SyntaxError: invalid syntax

6) PEP-8 (Python Enhancement Proposal version 8) document that provides guidelines and how to practice on how to write python code.

How to check whether the given python identifier is valid or not valid?

"identifier name".isidentifier()

eg: "abc".isidentifier()